CO2 – ASSESSMENT 1

REPORT ANALYSIS

Course: Ethical Hacking **Course Code:** ITA1406
Slot: Slot B **Total Marks:** 100

Topic: Domain & Email Footprinting using theHarvester

CO2: Footprinting Tools: Conducting Competitive Intelligence, DNS Zone Transfers; Introduction to Social Engineering: Social Engineering Attacks and Countermeasures (BT3).

Academic integrity note: This report includes explanations, commands, and illustrative terminal images. The supplied assignment contains no actual scan results, and no live scan was run here. Therefore, real email addresses, subdomains, hosts, IPs, and totals are not invented. The example screenshots are clearly marked as illustrative.

Objective

To understand how theHarvester collects publicly available domain-related information—such as email addresses, hostnames, subdomains, and IP addresses—from configured public sources, and to document and interpret results responsibly.

1. Tool installation and verification

The assignment specifies installing Python, upgrading pip, installing theHarvester, checking its help output, and running a scan for saveetha.com. Commands below follow that workflow. Installation methods may vary by operating system and current package version.

Windows Command Prompt (example)

```
python --version
python -m pip install --upgrade pip
python -m pip install theHarvester
theHarvester --help
theHarvester -d saveetha.com -b all
```

If the command is not recognized, reopen Command Prompt after installation, confirm Python is on PATH, and consult the current theHarvester documentation. Some versions may require a supported environment or package manager rather than pip.

2. Command execution screenshot

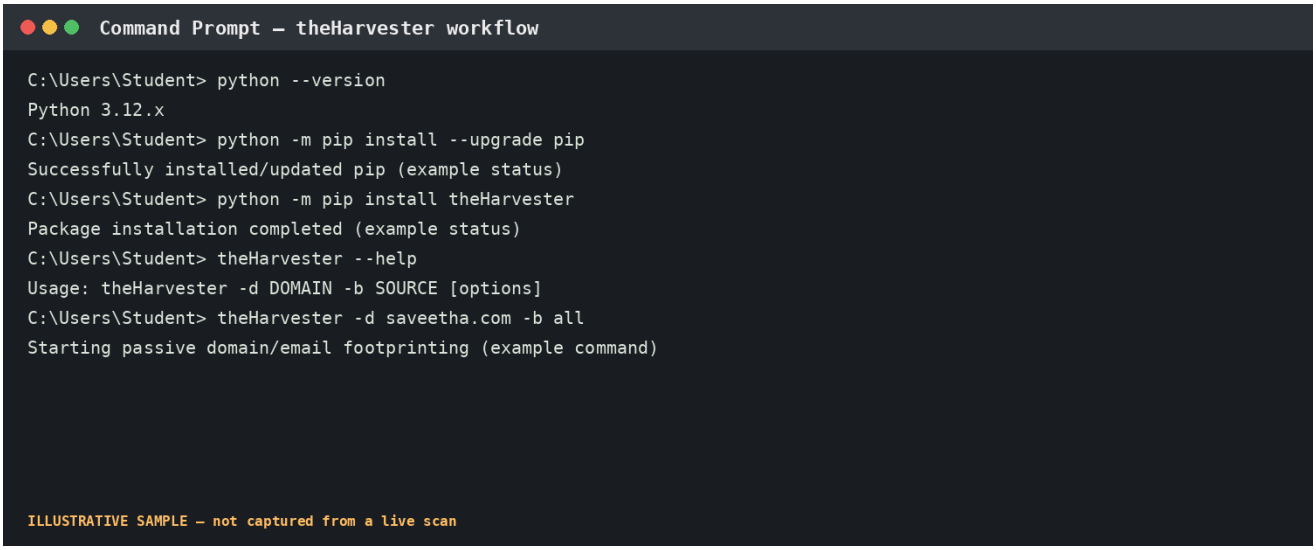


Figure 1. Illustrative command workflow only; not a screenshot of a real installation or scan.

3. Answers to the required fields

Required field	Answer / report entry
Domain Name	saveetha.com
Emails found (list one by one)	Not verifiable from the supplied question paper. Actual addresses require a live scan.
Sub emails found (list one by one)	Not verifiable from the supplied question paper. Record subdomains from actual tool output; do not in
Hosts and its IP (list one by one)	Not verifiable from the supplied question paper. Record each hostname and resolved IP exactly as re
Total Email Addresses Found	Unavailable without a live scan. Count unique email addresses in actual output.
Total Subdomains Found	Unavailable without a live scan. Count unique subdomains in actual output.
Total Hosts Found	Unavailable without a live scan. Count unique hostnames in actual output.
Total IP Addresses Found	Unavailable without a live scan. Count unique IP addresses in actual output.

4. theHarvester output screenshot

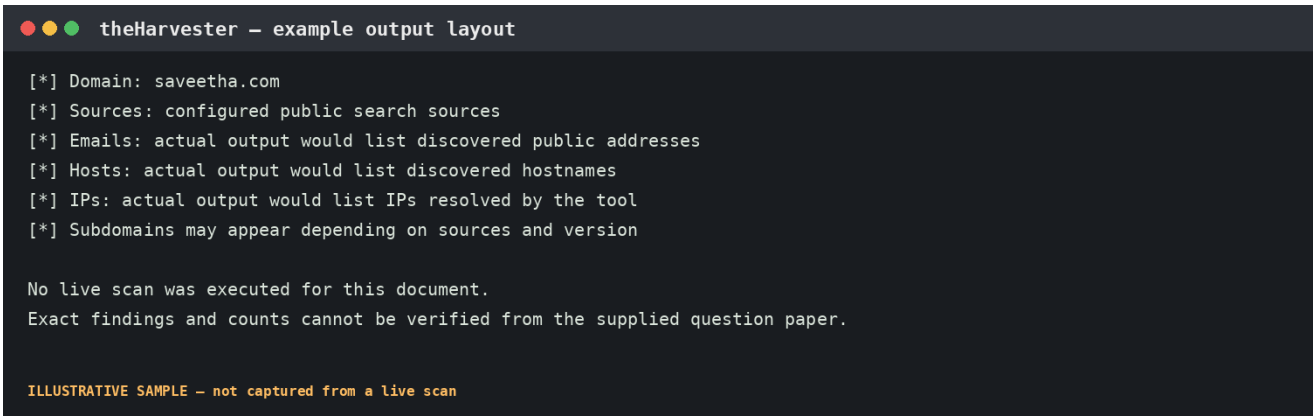


Figure 2. Illustrative output layout only. No fabricated addresses, hostnames, IP addresses, or totals are shown.

5. Analysis and interpretation

• Email addresses can reveal publicly indexed contact points associated with a domain and must be handled responsibly.
• Subdomains and hostnames can indicate public-facing services. Their presence alone does not prove a vulnerability.
• IP addresses may relate hostnames to network endpoints, but results can vary due to DNS changes, CDNs, shared hosting, and source coverage.
• theHarvester gathers information from configured sources; results depend on tool version, source availability, rate limits, and network access.
• A missing result does not prove that no email, host, or subdomain exists; it may simply not be indexed by the sources queried.

6. Result

The workflow for domain and email footprinting with theHarvester is documented, including installation and verification commands, the scan command, the required result fields, and interpretation guidance. Actual discovery lists and numerical totals remain unconfirmed because no live tool output was provided or executed.

7. Conclusion

This exercise demonstrates the purpose and basic workflow of passive footprinting. Accurate reporting requires retaining real command output, deduplicating findings, and entering only verified results. Reconnaissance should be limited to authorized and lawful educational use.

8. References

• theHarvester project: https://github.com/laramies/theHarvester
• Python downloads: https://www.python.org/downloads/
• Original assignment handout: CO2 – Assessment 1, Ethical Hacking, ITA1406, Slot B.

The original rubrics page from the supplied assignment is appended after this report.

RUBRICS

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (ProYicient)	Level 2 (Developing)	Level 1 (Beginning)
Tool Installation and ConYiguration	25%	Successfully installs, veriTies, and conTigures theHarvester and required environment without errors.	Installs and conTigures the tool correctly with minor issues.	Completes installation with assistance and encounters several conTiguration issues.	Unable to install, conTigure, or verify the tool.
Footprinting Data Collection	25%	Accurately collects and organizes emails, sub-emails, hosts, IP addresses, and subdomains comprehensively.	Collects most required footprinting information with minor omissions.	Collects limited information or misses several required Tields.	Fails to obtain meaningful footprinting information.
Analysis and Interpretation of Results	30%	Thoroughly analyzes theHarvester output and accurately interprets the signiTicance of discovered information.	Correctly interprets most results with minor errors.	Provides basic interpretation with limited technical understanding.	Unable to interpret the collected footprinting information.
Evidence and Documentation	20%	Provides complete screenshots, counts, lists, and organized documentation with technically accurate presentation.	Provides adequate evidence and documentation with minor omissions.	Documentation is partially complete or poorly organized.	Evidence is missing, incomplete, or difTicult to verify.